Ty Snyder

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EDUCATION

University of California, Berkeley May 2028
Bachelor of Arts in Data Science, Dual Minor in Mechanical Engineering and EIS GPA: 3.44

Relevant Coursework:
Foundations of Signals, Circuits & Devices, Solid Mechanics, Dynamic Systems & Feedback Controls, Data Structures, Principles & Techniques of Data Science, Electronics for the Internet of Things, Physics for Engineers, Intro to Machine Learning, Optimization Models

WORK EXPERIENCE

Sunday Robotics Redwood City, CA
Hardware Engineering Intern June 2026 – August 2026
- Modified the **entire fleet** with improved zeroing jigs and strain relief, enhancing reliability.
- Built an actuator testing jig to flash and validate 20 actuators at once, dramatically cutting preparation time.
- Wrote 2 SOP instruction sets, standardizing the build process, which reduced one off mistakes.

Intech Mechanical Roseville, CA
BIM Intern and Addon Programmer January 2023 – May 2026
- Developed a **C# 30k** line addon that optimized workflow, removing ~60% of repetitive Revit tasks.
- Produced detailed drawings for the field and shop teams for manufacturing and installation.
- Improved broken addons to save costs, make them more user-friendly, and more performant.

LEADERSHIP & INVOLVEMENT

Combat Robotics at Berkeley Berkeley, CA
President June 2026 – Present
- Manage a **$45k budget** and timelines to keep 120+ members on track for 6+ competitions.
- Assist returning members in idea generation, team forming, and the manufacturing of robots.
- Optimized drive on a 30lb combat robot to cancel gyroscopic effect while still having 11m/s max speed.
- Built a timing belt calculator to ensure belts fit well and prevent slipping.

OpenCal Research Lab Berkeley, CA
Undergrad Researcher August 2025 – Present
- Led CAD organization to keep the document running smoothly and user-friendly for public release.
- Designed a XYZ linear stage for fine adjustment capable of a 4lb projector load for 15$.
- Collaborated on designing an open-source Computation Axial Lithography (CAL) 3d printer.
- Optimized parameterization to be able to accommodate different projectors without full reCAD.

PROJECTS

Quadruped Robot V1 and V2 March 2023 – Present
Personal Project
- Derived from scratch inverse kinematics formula to send the feet to the correct end effector.
- Manufactured custom 3d printed chassis, which is able to hold its own weight with a FoS of 3.
- Programmed a Raspberry Pi to read a controller input via Bluetooth to control the quadruped.

SKILLS

Software: Autodesk Fusion, Onshape, Autodesk Workflow Automation, Revit, C#, Python, SQL, Java
Communication: Student Coaching, Team Leadership, Technical Documentation, Public Speaking
Technical: Mechanical Design, HVAC & Plumbing Systems, Sensors, Robotics, Automation, DFM